Note 4.1 The Mean Value Theorem

Extreme Values of Functions

- **Definition of Absolute Maximum and Minimum:**
  C is the number in the domain of $f(x)$. If for all $x \in D$, $f(x) \geq f(c)(f(x) \leq f(c))$, then $f(c)$ is called the absolute minimum(maximum) of f on D.
  Sometimes we call them global maximum and minimum. The maxima and minima are called extrema of f.
- **Definition of Relative Maximum and Minimum:**
  C is the number in the domain of $f(x)$. If when x is near $c, f(x) \geq f(c)(f(x) \leq f(c))$, then $f(c)$ is called a relative minimum(maximum) of f on D.
- **Fermat's Theorem:**
  If $f(x)$ has a local maximum or minimum at c and $f'(c)$ exist, then

$$f'(c) = 0$$

- **Rolle's Theorem:**
  Let $f(x)$ be a function that satisfies the following three hypotheses:
  - $f(x)$ is continuous on $[a, b]$.
  - $f(x)$ is differential on (a, b).
  - $f(a) = f(b)$
    Then there is a number c in (a, b) such that $f'(c) = 0$.
    Proof:
    Since $f(x)$ is continuous on $[a, b]$, then there exist a maximum M and m in $[a, b]$.
    If $M = m$, then $f(x)$ is a constant function on $[a, b]$. So for any point in $[a, b]$, $f'(x) = 0$.
    If $M \neq m$, since $f(a) = f(b)$, there exist at least one extrema in (a, b). Then by Fermat's Theorem, the point c that take the extrema satisfies $f'(c) = 0$.

The Mean Value Theorem

- **Lagrange's Mean Value Theorem:**
  If $f(x)$ is continuous on $[a, b]$ and differentiable on (a, b), then there exists a number c in (a, b) such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

or equivalently,

$$f(b) - f(a) = f'(c)(b - a)$$

- **Proof:**
  Idea:

$$f'(c) = \frac{f(b)-f(a)}{b-a} \Leftrightarrow f'(c) - \frac{f(b)-f(a)}{b-a} = 0 \Leftrightarrow \left(f(x) - \frac{f(b)-f(a)}{b-a}x\right)' = 0 \Leftrightarrow g'(x) = 0$$

  where $g(x) = f(x) - \frac{f(b)-f(a)}{b-a}$.
  then

$$g(a) = f(a) - \frac{af(b)-af(a)}{b-a} = \frac{bf(a)-af(b)}{b-a}$$

$$g(b) = f(b) - \frac{bf(b)-bf(a)}{b-a} = \frac{bf(a)-af(b)}{b-a}$$

  thus $g(a) = g(b)$. By Rolle's Theorem, there exist a $c \in (a,b)$ such that $g'(c) = 0$
- **Corollary 1:**
  If $f'(x) = 0$ for all x in (a,b), then $f(x)$ is constant on (a,b).
  Proof:
  Suppose arbitrary x_1, x_2 are in (a,b), then by Lagrange's Mean Value Theorem, there exists a point c such that

$$f'(c) = \frac{f(x_2)-f(x_1)}{x_2-x_1} = 0$$

  then we get

$$f(x_1) = f(x_2)$$

  So $f(x)$ is constant on (a,b).
- **Corollary 2:**
  If $f'(x) = g'(x)$ at each point in the interval (a,b), then there exists a constant C such that $f(x) = g(x) + C$ for all x in (a,b). That is, $f-g$ is a constant function on (a,b).
  Proof:
  Let $h(x) = f(x) - g(x)$ on the interval (a,b), then by Lagrange's Mean Value Theorem, there exist some c such that

$$h'(c) = \frac{h(b)-h(a)}{b-a}$$

  Since $f'(x) - g'(x) = 0$, then $h'(c) = 0$. Thus $h(b) - h(a) = 0$. Then $h(x) = C$ on (a,b). Thus $f(x) = g(x) + C$.